

# The 168 Quantum of Non-Associativity in the Cayley–Dickson Tower

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**Abstract.** We study the number  $T_k$  of ordered triples of imaginary basis elements with nonzero associator in the  $2^k$ -dimensional Cayley–Dickson algebra. It is classical that  $T_3 = 168 = |\mathrm{PSL}(2, 7)|$  for the octonions. By exhaustive computation we establish  $T_4 = 1848$ ,  $T_5 = 15\,960$ , and  $T_6 = 130\,200$ , and observe that  $T_k = 168(P_k - 4P_{k-1})$  where  $P_k = (2^k - 1)(2^{k-1} - 1)(2^{k-2} - 1)/21$  counts Fano subplanes in  $\mathrm{PG}(k - 1, 2)$ . This formula is verified at four levels ( $k = 3, 4, 5, 6$ ). We derive that  $T_k/(2^k - 1)^3 \rightarrow 1/2$  as  $k \rightarrow \infty$ : asymptotically, exactly half of all basis triples are non-associative. The associator norm dichotomy  $\|[e_i, e_j, e_k]\| \in \{0, 2\}$  — previously known for octonions — is verified exhaustively through dimension 64. The number of Fano subplanes grows by a factor of 7 per Cayley–Dickson doubling, yielding a combinatorial scaling exponent  $\log_2 7 \approx 2.807$  for the non-associative structure.

**Keywords:** Cayley–Dickson algebras, non-associativity, Fano plane, projective geometry,  $\mathrm{PSL}(2, 7)$

**MSC 2020:** 17A75, 20B25, 17D05, 51E20

## 1. Introduction

The Cayley–Dickson construction produces a tower of real algebras by iterated doubling:

$$\mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow \mathbb{S} \rightarrow \mathbb{T} \rightarrow \dots \quad (1)$$

At each step, a structural property is lost: commutativity at the quaternions  $\mathbb{H}$ , associativity at the octonions  $\mathbb{O}$ , and the division property at the sedenions  $\mathbb{S}$  [1, 2]. In a companion paper [3], we proved that the number of ordered triples  $(i, j, k) \in \{1, \dots, 7\}^3$  of imaginary octonion basis elements with nonzero associator is exactly  $T_3 = 168 = |\mathrm{PSL}(2, 7)|$ , following from the regular action of  $\mathrm{Aut}(\mathrm{Fano})$  on ordered non-collinear triples.

A natural question arises: *does the 168-divisibility persist in higher Cayley–Dickson algebras?* In the sedenions ( $2^4 = 16$  dimensions,  $15^3 = 3375$  basis triples) and beyond, new phenomena appear — zero divisors, non-alternativity, and a proliferation of Fano subplanes in the associated projective geometry  $\mathrm{PG}(k - 1, 2)$ . The nonzero associator count  $T_k$  grows rapidly, but its relationship to 168 has not been studied.

In this paper we address this gap. By exhaustive computation through  $k = 6$  (dimension 64, requiring classification of  $63^3 = 250\,047$  triples), we establish that  $T_k$  is a multiple of 168 at every level tested, and conjecture a closed formula relating  $T_k$  to the number of Fano subplanes in  $\mathrm{PG}(k - 1, 2)$ . We derive the

asymptotic fraction of non-associative triples and identify a combinatorial scaling exponent for the Fano subplane structure.

### Contributions.

1.  $T_k$  computed exhaustively for  $k = 3, 4, 5, 6$  (Theorem 1).
2. A conjectured closed formula  $T_k = 168(P_k - 4P_{k-1})$  verified at four levels (Conjecture 2).
3. Asymptotic analysis:  $T_k/N^3 \rightarrow 1/2$  (Proposition 3).
4. The binary norm property  $\|[e_i, e_j, e_k]\| \in \{0, 2\}$  verified through dimension 64 (Observation 4).
5. Fano subplane branching exponent  $\log_2 7 \approx 2.807$  (Proposition 5).

## 2. Preliminaries

### 2.1. Cayley–Dickson algebras

Given a real algebra  $A$  with conjugation  $\bar{\cdot}$ , the Cayley–Dickson doubling defines a new algebra  $A' = A \times A$  with multiplication:

$$(a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c}) \quad (2)$$

Starting from  $\mathbb{R}$ , this yields  $\mathbb{C}$ ,  $\mathbb{H}$ ,  $\mathbb{O}$ ,  $\mathbb{S}$ , and the  $2^k$ -dimensional algebra  $A_k$  for each  $k \geq 0$ . The algebra  $A_k$  has  $2^k - 1$  imaginary basis elements  $e_1, \dots, e_{2^k-1}$ , with  $e_0 = 1$  the identity. The associator of three elements is  $[a, b, c] = (ab)c - a(bc)$ .

### 2.2. Fano subplanes in $\text{PG}(k-1, 2)$

The imaginary basis elements of  $A_k$  are naturally indexed by nonzero vectors of  $\mathbb{F}_2^k$ , and the projective space  $\text{PG}(k-1, 2)$  has  $2^k - 1$  points. A **Fano subplane** is a subgeometry isomorphic to  $\text{PG}(2, 2)$  — the Fano plane, with 7 points and 7 lines. The number of Fano subplanes in  $\text{PG}(k-1, 2)$  is:

$$P_k = \frac{(2^k - 1)(2^{k-1} - 1)(2^{k-2} - 1)}{21} \quad (3)$$

with  $P_2 = 0$ ,  $P_3 = 1$ ,  $P_4 = 15$ ,  $P_5 = 155$ ,  $P_6 = 1395$  [4].

## 3. Results

### 3.1. Exhaustive computation of $T_k$

We define  $T_k$  as the number of ordered triples  $(i, j, k)$  with  $i, j, k \in \{1, \dots, 2^k - 1\}$  such that  $[e_i, e_j, e_k] \neq 0$  in  $A_k$ .

**Theorem 1.** *The nonzero basis associator counts for  $k = 3, 4, 5, 6$  are:*

| $k$ | Dim | $N = 2^k - 1$ | $N^3$   | $T_k$   | $T_k/168$ |
|-----|-----|---------------|---------|---------|-----------|
| 3   | 8   | 7             | 343     | 168     | 1         |
| 4   | 16  | 15            | 3 375   | 1 848   | 11        |
| 5   | 32  | 31            | 29 791  | 15 960  | 95        |
| 6   | 64  | 63            | 250 047 | 130 200 | 775       |

*Each value is verified by exhaustive enumeration of all  $N^3$  ordered triples.*

For  $k = 3$ , the result is classical (168 = number of ordered non-collinear triples in the Fano plane [1]). For  $k = 4$ , the value  $T_4 = 1848 = 11 \times 168$  was reported in [3]. The values  $T_5$  and  $T_6$  are new.

The computation for  $k = 6$  requires classifying 250,047 ordered triples in a 64-dimensional algebra. Each associator computation involves four Cayley–Dickson multiplications (recursively through 6 levels of doubling). Two independent runs produced identical results.

### 3.2. A conjectured closed formula

**Conjecture 2.** *For all  $k \geq 3$ :*

$$T_k = 168(P_k - 4P_{k-1}) \quad (4)$$

where  $P_k$  is given by Equation 3.

**Verification.** The conjecture is confirmed at all four computed levels:

| $k$ | $P_k$ | $P_{k-1}$ | $168(P_k - 4P_{k-1})$       | Match |
|-----|-------|-----------|-----------------------------|-------|
| 3   | 1     | 0         | $168 \times 1 = 168$        | ✓     |
| 4   | 15    | 1         | $168 \times 11 = 1848$      | ✓     |
| 5   | 155   | 15        | $168 \times 95 = 15\,960$   | ✓     |
| 6   | 1395  | 155       | $168 \times 775 = 130\,200$ | ✓     |

The factor  $P_k - 4P_{k-1}$  admits a structural interpretation. Each Fano subplane in  $\text{PG}(k-1, 2)$  supports a copy of the octonion multiplication table. In the Cayley–Dickson algebra  $A_k$ , these subplanes split into two classes: *octonionic* subplanes (where the restricted algebra is alternative, contributing 168 nonzero associator triples each) and *quasi-octonionic* subplanes (where alternativity fails, contributing fewer). The structure of this decomposition has been studied for sedenions by Cawagas [5]. A proof of Equation 4 for general  $k$  would require establishing that quasi-octonionic subplanes contribute exactly 72 nonzero triples each and that their count is  $7P_{k-1}$ . We leave this as an open problem.

### 3.3. Asymptotic analysis

**Proposition 3.** *If Conjecture 2 holds, then  $T_k/(2^k - 1)^3 \rightarrow 1/2$  as  $k \rightarrow \infty$ .*

*Proof.* For large  $k$ ,  $P_k \approx 2^{3k-3}/21$  (from Equation 3). Then:

$$T_k \approx 168 \cdot (2^{3k-3}/21 - 4 \cdot 2^{3(k-1)-3}/21) = 168 \cdot 2^{3k-6} \cdot (8 - 4)/21 = 168 \cdot 4 \cdot 2^{3k-6}/21 = 2^{3k-1} \quad (5)$$

Since  $(2^k - 1)^3 \approx 2^{3k}$ , the ratio  $T_k/(2^k - 1)^3 \rightarrow 2^{3k-1}/2^{3k} = 1/2$ .  $\square$

This means that asymptotically, *exactly half* of all ordered basis triples are non-associative. The convergence is visible from the data: the fraction is 0.490 at  $k = 3$ , 0.548 at  $k = 4$ , 0.536 at  $k = 5$ , and 0.521 at  $k = 6$ , approaching 0.500 from above.

### 3.4. Norm dichotomy

**Observation 4.** *For  $k = 3, 4, 5, 6$ , the associator norm on imaginary basis elements satisfies*

*$\| [e_i, e_j, e_k] \| \in \{0, 2\}$ . No intermediate values are observed.*

For octonions ( $k = 3$ ), this is a consequence of alternativity [3]. For  $k \geq 4$ , alternativity fails, and the persistence of the  $\mathbb{Z}_2$  dichotomy is not a priori guaranteed. The exhaustive verification at  $k = 6$  (250,047 triples, all norms exactly 0 or 2) suggests this is a general property of the Cayley–Dickson construction, likely derivable from the bilinearity of the structure constants and the  $\pm 1$  character of the Cayley–Dickson signs.

### 3.5. Fano subplane branching exponent

**Proposition 5.** *If Conjecture 2 holds and the quasi-octonionic plane count is  $Q_k = 7P_{k-1}$ , then the total number of Fano subplanes satisfies  $P_k/P_{k-1} \rightarrow 2^3 = 8$  as  $k \rightarrow \infty$ , and the number of non-associative triples per Fano subplane scales with branching factor 7, yielding a combinatorial scaling exponent of  $\log_2 7 \approx 2.807$ .*

*Proof.* From  $P_k \approx 2^{3k-3}/21$ , we have  $P_k/P_{k-1} \approx 8$ . Each doubling creates  $P_k - P_{k-1} \approx 7P_{k-1}$  new quasi-octonionic planes (and the  $P_{k-1}$  existing planes persist as octonionic). The branching factor for the non-octonionic structure is therefore 7, and the scaling exponent in the logarithmic base of the dimension ( $2^k$ ) is  $\log_2 7 \approx 2.807$ .  $\square$

The codimension of the Fano subplane structure in the three-dimensional triple space is  $3 - \log_2 7 \approx 0.193$ . This means the non-associative structure, while filling asymptotically half the triple space by measure, is organized around a Fano subplane skeleton of sub-integer dimension.

## 4. Computational methods

All computations use Cayley–Dickson multiplication implemented recursively:  $(a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c})$ , building from real arithmetic through  $k$  levels of doubling. For  $k = 6$ , each multiplication involves  $2^6 = 64$  coordinates and recurses through 6 levels.

The  $k = 5$  computation (29,791 triples) requires approximately 2 minutes. The  $k = 6$  computation (250,047 triples) requires approximately 57 minutes. Two independent implementations produced identical results for all values of  $k$ .

Source code is available at [github.com/agourakis82/sounio](https://github.com/agourakis82/sounio).

## 5. Open questions

1. **Prove Conjecture 2 for all  $k \geq 3$ .** The key step is establishing the octonionic/quasi-octonionic subplane decomposition: show that each Fano subplane in  $A_k$  is either octonionic (contributing 168 nonzero associator triples) or quasi-octonionic (contributing 72), and that the quasi-octonionic count is  $7P_{k-1}$ .
2. **Prove that  $\| [e_i, e_j, e_k] \| \in \{0, 2\}$  for all  $k$ .** The norm dichotomy is verified through  $k = 6$  but not proven in general. A proof may follow from the structure of the Cayley–Dickson sign function.
3. **Compute  $T_7$ .** The formula predicts  $T_7 = 168 \times 6231 = 1\,046\,808$ . Exhaustive verification would require  $127^3 \approx 2 \times 10^6$  associator evaluations in dimension 128, which is computationally feasible.
4. **Characterize the sub-class distribution.** For sedenions, the nonzero associator triples decompose by index range (oct-oct-oct, sed-sed-sed, cross) with each sub-count a multiple of 168 [3]. Does this hold for  $k \geq 5$ ?

## 6. Acknowledgments

The octonion base result ( $T_3 = 168$ ) was established in joint work with M. Gerenutti [3].

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